Number Sequences (Set B)

Year 4 Pt 2 · **Lesson Worksheet**

Worked Examples

Example 1 — Arithmetic Sequence: What comes next? 4, 9, 14, 19, __

Answer: 24 — each term increases by 5 (add 5 each time).

Example 2 — Geometric Sequence: What comes next? 2, 6, 18, 54, __

Answer: 162 — each term is multiplied by 3.

Example 3 — Two-Step Rule: What comes next? 1, 3, 7, 15, 31, __

Answer: 63 — each term is double the previous term, plus 1 (×2, +1).

Spotting the Pattern (15 questions)

1. What comes next? 3, 6, 9, 12, __

2. What comes next? 50, 45, 40, 35, __

3. What comes next? 2, 4, 8, 16, __ A) 20 B) 24 C) 28 D) 32 E) 36

4. Select ALL that apply: which of these sequences increase by exactly 4 each time? A) 2, 6, 10, 14 B) 5, 9, 13, 18 C) 1, 5, 9, 13 D) 3, 7, 12, 16 E) 10, 14, 18, 22

5. What comes next? 7, 14, 21, 28, __

6. What comes next? 100, 90, 80, 70, __ A) 50 B) 55 C) 60 D) 65 E) 75

7. What comes next? 1, 4, 9, 16, 25, __ (these are square numbers)

8. Fill in the missing number: 5, 10, __, 20, 25 (increasing by 5 each time)

9. What comes next? 1, 2, 4, 8, 16, __ A) 20 B) 24 C) 28 D) 32 E) 36

10. Select ALL that apply: which of these are true about the sequence 3, 9, 27, 81? A) Each term is multiplied by 3. B) Each term increases by 6. C) The next term would be 243. D) The sequence is geometric. E) The sequence is arithmetic.

11. What comes next? 2, 5, 8, 11, __

12. What comes next? 1, 1, 2, 3, 5, 8, __ (the Fibonacci sequence — add the two previous terms) A) 11 B) 12 C) 13 D) 14 E) 15

13. What comes next? 60, 54, 48, 42, __

14. Fill in the missing number: 2, 4, 8, __, 32 (each term doubles)

15. Select ALL that apply: which statements about the sequence 1, 4, 9, 16, 25 are true? A) These are the first five square numbers. B) The next term is 36. C) Each term increases by a constant amount. D) The gap between terms increases each time. E) The sequence is decreasing.

Applying the Pattern (15 questions)

1. What comes next? 4, 7, 12, 19, 28, __ (the gap increases by 2 each time: +3, +5, +7, +9…)

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2. What comes next? 100, 50, 25, 12.5, __

3. What comes next? 2, 3, 5, 8, 13, __ (Fibonacci-style: add the two previous terms)

4. A sequence follows the rule "double, then subtract 1". Starting at 3, what is the 4th term? (3, 5, 9, __)

5. Select ALL that apply: which of these sequences follow a Fibonacci-style rule (each term is the sum of the two before it)? A) 1, 1, 2, 3, 5, 8 B) 2, 4, 6, 8, 10 C) 2, 2, 4, 6, 10, 16 D) 1, 2, 3, 4, 5, 6 E) 3, 5, 8, 13, 21, 34

6. What comes next? 5, 8, 14, 23, 35, __ (the gap increases by 3 each time: +3, +6, +9, +12...)

7. A sequence follows the two-step rule "multiply by 2, then add 3". Starting at 1, what are the first four terms?

8. What comes next in this sequence of triangular numbers? 1, 3, 6, 10, 15, __

9. In the sequence 3, 6, 11, 18, 27, __, what is the rule linking consecutive terms, and what comes next?

10. Select ALL that apply: which of these sequences have a gap that increases by the same amount each time (i.e. the gaps themselves form an arithmetic sequence)? A) 1, 2, 4, 7, 11 (gaps 1,2,3,4) B) 2, 4, 6, 8, 10 (gaps all 2) C) 1, 3, 6, 10, 15 (gaps 2,3,4,5) D) 5, 10, 20, 40 (gaps 5,10,20) E) 2, 5, 9, 14, 20 (gaps 3,4,5,6)

11. What comes next? 200, 100, 50, 25, __

12. A sequence starts 2, 5 and each new term is found by adding the previous two terms. What are the next three terms after 2, 5?

13. What comes next? 1, 2, 6, 24, 120, __ (hint: each term is multiplied by an increasing amount: ×2, ×3, ×4, ×5...)

14. Select ALL that apply: which of these are geometric sequences (each term multiplied by the same constant ratio)? A) 4, 8, 16, 32 B) 3, 6, 9, 12 C) 100, 50, 25, 12.5 D) 2, 6, 18, 54 E) 1, 3, 5, 7

15. A sequence follows the rule "subtract 2, then multiply by 2". Starting at 10, find the next two terms after 10.

Sequences in Context (15 questions)

1. What comes next in this sequence, and explain the two-step rule? 2, 5, 11, 23, 47, __

2. A sequence is defined by: term = (previous term × 2) − 3. If the first term is 4, find the first five terms of the sequence.

3. Select ALL that apply: for the sequence 1, 2, 4, 7, 11, 16, __, which of these are correct? A) The gaps are 1, 2, 3, 4, 5... B) The next term is 22. C) The sequence is geometric. D) The gap increases by 1 each time. E) The 7th term is 23.

4. A sequence of numbers is built from two interleaved patterns: the odd-position terms are 1, 3, 5, 7... and the even-position terms are 2, 6, 10, 14... Write out the first eight terms of the combined sequence.

5. What comes next, and what is the rule? 3, 4, 7, 11, 18, 29, __ (hint: compare to the Fibonacci idea, but check the starting numbers carefully)

6. A sequence starts at 100 and each term is found by halving the previous term and then adding 1. Find the first five terms, starting with 100.

7. Select ALL that apply: a student claims the sequence 1, 8, 27, 64, __ continues with 100. Which of these statements are correct? A) These are cube numbers (1^3, 2^3, 3^3, 4^3...). B) The next term should be $5^3 = 125$, not 100. C) 100 is correct because the gaps are increasing. D) The gaps are 7, 19, 37, which do not increase by a constant amount, ruling out a simple arithmetic explanation. E) The sequence is arithmetic.

8. A sequence follows "add the term's position number" — so term 1 is 5, and each new term adds the NEXT position number (term 2 = term 1 + 2, term 3 = term 2 + 3, and so on). Find the first five terms starting from 5.

9. Explain why the sequence 2, 6, 12, 20, 30 is NOT a simple arithmetic or geometric sequence, then identify the actual rule and give the next term.

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10. What comes next in this sequence, and explain the rule, which involves TWO alternating operations: 3, 9, 7, 13, 11, __ (hint: alternately add 6 and subtract 2)

11. A sequence is formed by adding consecutive odd numbers to a running total, starting at 0: 0, +1, +3, +5, +7… giving totals 0, 1, 4, 9, 16… Explain what special numbers these totals are, and give the 6th total.

12. Select ALL that apply: for the sequence 80, 40, 20, 10, 5, __, which of the following are true? A) Each term is half the previous term. B) The next term is 2.5. C) This is an arithmetic sequence. D) This is a geometric sequence. E) The common ratio is 1/2.

13. A sequence follows the rule: term(n) = term(n−1) + term(n−2) − 1, starting with 2 and 3. Find the next three terms after 2, 3.

14. What comes next, and explain the rule? 1, 2, 4, 7, 12, 20, __ (hint: look at how each gap relates to the PREVIOUS gap, not to a fixed amount)

15. A sequence of numbers represents the number of handshakes needed if everyone in a group shakes hands with everyone else once: for groups of 2, 3, 4, 5 people, the totals are 1, 3, 6, 10. Identify the pattern type and predict the total for a group of 6 people.

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